TRESSE
vāld.labs

USER MANUAL
V1.0.0 · MAY 2026

// 8-VOICE POLYPHONIC MULTI-ENGINE SYNTHESIZER

The Art of Shaping Chaos.

Eight voices, **thirty-nine** ways to make them sing, and a SEM filter that morphs LP → BP → NOTCH → HP under one knob. This is the **field guide** — every knob the plugin has, in the order you'll reach for it.

ALPHA **Public alpha build.** Most of the surface is locked, but a few engine parameters and the preset browser are still settling. Bug reports and wishlist items: **vald.labs.lisbon@gmail.com**.

MACOS · AU · VST3 · STANDALONE

DESIGNED WITH LOVE
BY **VALD LABS** · LISBON

// 00 — CONTENTS

What's in here.

Fifteen short chapters. The first four get you playing in a minute. The rest are the corners — voice allocation, the filter morph, every one of the thirty-nine engines — the bits you'll lean on once you stop being polite to the plugin.

// GET GOING

// THE SOUND

// MECHANICS

// MUSICAL

// HOUSEKEEPING

// RECIPES

// 01 — INSTALL & FIRST BOOT

Run the PKG. Open your DAW.

TRESSE ships as a universal macOS package. The installer drops the VST3 into the path your DAW already scans, and the AU alongside it. There's a Standalone app in the same bundle if you'd rather not boot a host.

IF MACOS BLOCKS THE INSTALLER

The PKG is ad-hoc signed, which Gatekeeper distrusts on first run. The workflow is two steps — try to open it, then tell macOS you meant it.

01 **Double-click** `TRESSE-1.0.0.pkg`. A dialog appears saying it can't be opened. Click **Done** — don't move it to Trash, that defeats the next step.

02 **Open System Settings → Privacy & Security.** Scroll to the **Security** section near the bottom.

03 You'll see a line reading **"TRESSE-1.0.0.pkg was blocked to protect your Mac."**

04 Click **Open Anyway**. Authenticate with your password or Touch ID.

05 The blocked-app dialog reappears, this time with an **Open** button. Click it — the installer runs.

WHERE THE PLUGIN LIVES

```
// installed by the PKG
/Library/Audio/Plug-Ins/VST3/TRESSE.vst3
/Library/Audio/Plug-Ins/Components/TRESSE.component
```

IF YOUR DAW REJECTS IT

macOS sometimes quarantines plugins copied across machines or downloaded twice. Strip the flag and it loads:

```
// in Terminal
sudo xattr -dr com.apple.quarantine \
  /Library/Audio/Plug-Ins/VST3/TRESSE.vst3
```

FIRST BOOT

- Drop TRESSE on an instrument track.
- Play some notes from your DAW.
- Tap `RANDOM` in the header until something catches your ear.
- Lock what you love by saving a preset, then keep going.

QUICK ORIENTATION

If the panel looks loud, that's the point. Twenty-four knobs and they're all wired to something audible. There are no decorative meters.

HEADS UP · OLDER MACOS

On Monterey and below the panel is called **Security & Privacy** and the **Open Anyway** button lives under a **General** tab. Same idea, different label.

// QUICK START — ABLETON

Ableton in thirty seconds.

Five steps to get a sound out. TRESSE is a regular instrument plugin — there's no routing trick, no second track. Drop, arm, play.

01 Drop TRESSE on a new MIDI track.

It registers as an instrument. No second track, no MIDI From routing — TRESSE makes the sound itself.

02 Arm the track so it monitors MIDI in.

Use the keyboard, your controller, or a clip. Whatever you're playing with.

03 Play a chord. Eight voices — you can lean on it.

If nothing comes out, check Monitor is set to **In** and your input channel is right.

04 Tap RANDOM in the header. New patch every time.

The current engine stays put. Only the 24 knobs reroll.

05 Pick an engine from the dropdown when you want to steer.

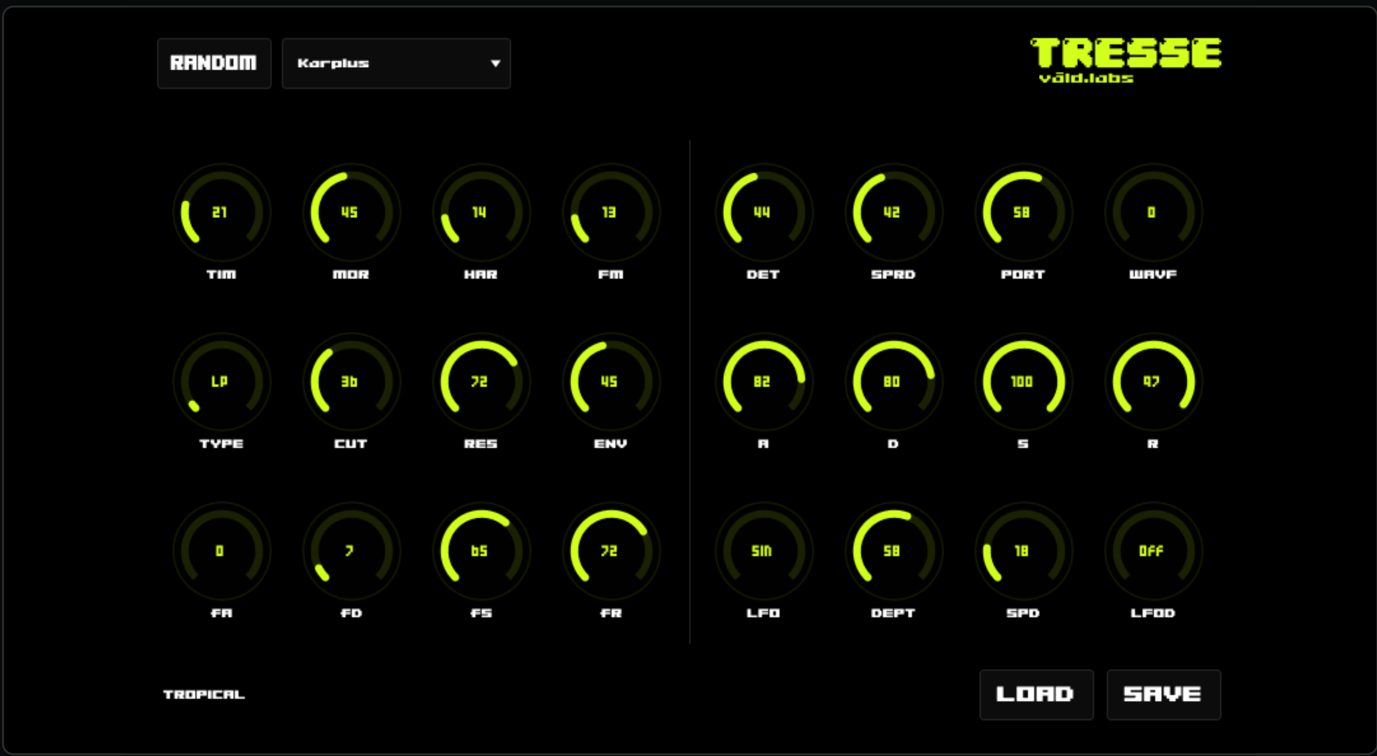
Thirty-nine of them. Chapter 11 is the cheat sheet.

MIDI → TRESSE → AUDIO

No bus. No bridge. Just a plugin. **That's the whole setup.**

// 02 — THE WINDOW, IN ONE LOOK

Header, two blocks, footer.



HEADER STRIP

Two buttons. `RANDOM` rerolls every knob on the panel; the engine dropdown picks one of 39 generators. The brand mark sits top-right.

THE TWO BLOCKS

Left = synthesis (TIM/MOR/HAR/FM) over the SEM filter (TYPE/CUT/RES/ENV) over the filter envelope (fA/fD/fS/fR). Right = voice movement, amp ADSR, LFO. A vertical hairline divides them — that's the signal-flow boundary.

FOOTER

Preset name on the left (currently `INIT` or whatever you loaded), `LOAD` and `SAVE` on the right for `.tresse` files.

> **READING THE PANEL**
>
> Row 1 of the left block is timbral. Row 2 is the filter. Row 3 is the filter envelope. The right block is voice + amp + LFO. Once that grid is in your head, you can stop reading the labels.

// 03 — HEADER

Two buttons.
Lazy or Pro choices here.

The header is the command palette. You'll spend most of your time on `RANDOM` — it's the dice — and reach for the engine dropdown when the dice land somewhere you want to explore.

RANDOM | Karplus ▾ | TRESSE vald.labs

RANDOM	The dice. Rerolls all 24 knobs at once. The engine stays put — so you keep the synthesis character and reshuffle everything around it. Lean on this. It's how the plugin wants to be played.
ENGINE	Picks one of 39 generators. Each engine reads TIM / MOR / HAR / FM differently — see chapter 11. Switching engines doesn't randomize anything; only the sound source changes.
PRESET	The footer holds `LOAD` and `SAVE` for `.tresse` files. Default name is `INIT`; loaded names are uppercased and shown to the left.

THE RANDOM HABIT

Pick an engine you like the personality of, then mash `RANDOM` until something musical falls out. Save what you love. Reroll the rest. It beats programming from a blank slate — and it's how every patch in the factory bank was started.

But navigating the engines gives you a deeper insight on the depth of the machine.

// 04 — SYNTHESIS BLOCK

Four knobs.
Thirty-nine personalities.

The top row of the left block is the soul of each engine. Four knobs — `TIM`, `MOR`, `HAR`, `FM` — and they mean something different for every engine on the list.

KNOB	FUNCTION	NOTES
TIM	Primary timbral control.	Cutoff, FM index, brightness, damping — the engine decides.
MOR	Secondary timbral control.	Resonance, waveform, algorithm select — the engine decides.
HAR	Harmonic content.	Chord type, partial spread, feedback, ratio — the engine decides.
FM	Internal FM amount.	Routes to each engine's modulation input.

WHY THE LABELS ARE SHORT

This is by design, not a bug. Some engines invert what feels intuitive — one might use `TIM` for cutoff and `MOR` for resonance, the next uses `TIM` for FM index and `MOR` for the algorithm. The cheat sheet is in chapter 11; better still, just turn the knobs and listen.

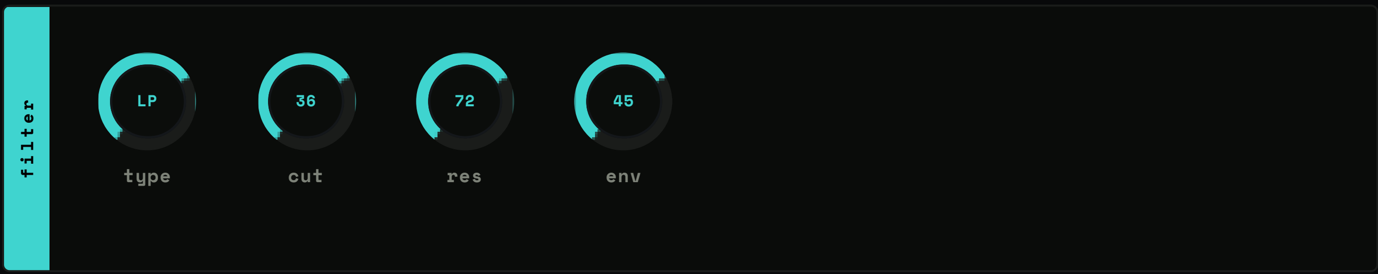
WHAT TO TRY FIRST

Hold a chord. Turn each knob to noon. Then sweep them one at a time and listen to what each one owns on the current engine. After two or three engines you'll start recognizing the patterns.

// 05 — THE SEM FILTER

One filter.
Four modes under a single knob.

Row 2 of the left block is a 2-pole state-variable filter built on a **TPT** (topology-preserving transform) structure. Resonance saturates inside the feedback loop, which means it can't blow up — turn RES to the ceiling and it self-stabilizes instead of screaming.



KNOB	RANGE
TYPE	Continuous mode morph — LP → BP → NCH → HP .
CUT	Cutoff frequency, 20 Hz → ~20 kHz, exponential.
RES	Resonance / Q, 0 → near self-oscillation, stable everywhere.
ENV	Filter envelope amount, 0–100 % modulation depth.

ABOUT THE MORPH

TYPE doesn't switch — it **morphs**. Sweep it from minimum to maximum and you cross continuously from low-pass through band-pass, through notch, into high-pass. The two endpoints of every adjacent mode pair blend through the middle. The classic Oberheim SEM character: one knob, infinite filter types between the named ones.

ABOUT THE RESONANCE

Inside the feedback path there's a tanh-style nonlinearity. It saturates the resonance before it can run away. The filter is self-stable at any Q without external clamping — so you can park RES at the top and play the cutoff like an instrument.

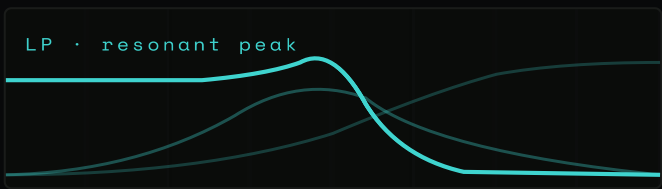


Fig. 02 — Same TYPE knob, three frequency responses at high RES. The peak rides the cutoff; tanh keeps it from clipping.

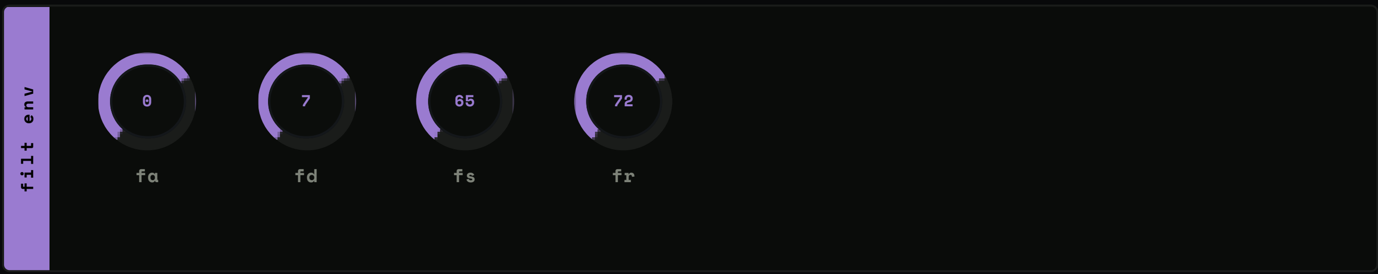
TRY THIS

Hold a sustained note. Park CUT around 60 %, push RES to the top, then sweep TYPE slowly from 0 to 1. That's the filter's whole personality in eight seconds.

// 06 — THE FILTER ENVELOPE

Attack, decay, sustain, release. For the cutoff alone.

Row 3 of the left block is a dedicated ADSR that drives the filter cutoff. Amount is set by ENV in the row above; positive at zero, opens upward as you turn it up. The amp envelope runs independently — they don't share a clock.



KNOB	RANGE
FA	Filter attack time, 1 ms → 4 s, exponential curve.
FD	Filter decay time, 1 ms → 4 s, exponential curve.
FS	Filter sustain level, 0–100 %.
FR	Filter release time, 1 ms → 4 s, exponential curve.

HOW IT TRIGGERS

Gate-on triggers the envelope. Gate-off enters release. The amp envelope runs **independently** on its own clock — a long filter release can outlive a short amp release, which is exactly the trick behind the classic plucky filter sound.

THE SHAPE

Fig. 03 — Cutoff travel scaled by ENV.

THE PLUCKY-FILTER RECIPE

Fast fA, medium fD, low fS, mid fR, with ENV around 60 % and RES around 70 % is the classic plucky filter sound. Five knobs, instant character. Works on any engine with harmonic content above the cutoff.

// 07 — VOICE MOVEMENT

Detune, spread, glide, fold.

Row 1 of the right block is where the eight voices stop sounding like clones of each other. Four knobs that turn unison into an ensemble.

DETUNE

Spreads the voices in pitch around the played note. At zero, eight voices in unison. At the top, ±25 cents — chorus, ensemble, supersaw-adjacent territory depending on the engine.

SPREAD

Pans the voices across the stereo field. At zero, everything's mono. At one, voices fan from full left to full right. Combine with DET and you've got an instant ensemble.

PORTAMENTO

Each new note starts at the previous note's pitch and glides to the target. The first note you play snaps — no opening glide from silence. Works in poly — each voice independently glides from the prior global note pitch.

WAVEFOLD

Post-filter wavefolder. Subtle warmth at low amounts, screaming harmonics at the top. Sits **after** the filter so the harmonics it generates are unfiltered — useful for adding bite without losing the filter character below it.

KNOB	RANGE
DET	Per-voice detune, 0 → ±25 cents.
SPRD	Stereo spread, 0 (mono) → 1 (full L/R).
PORT	Portamento, 0 (instant) → 1 (slow glide).
WAVF	Post-filter wavefold, 0–100 %.

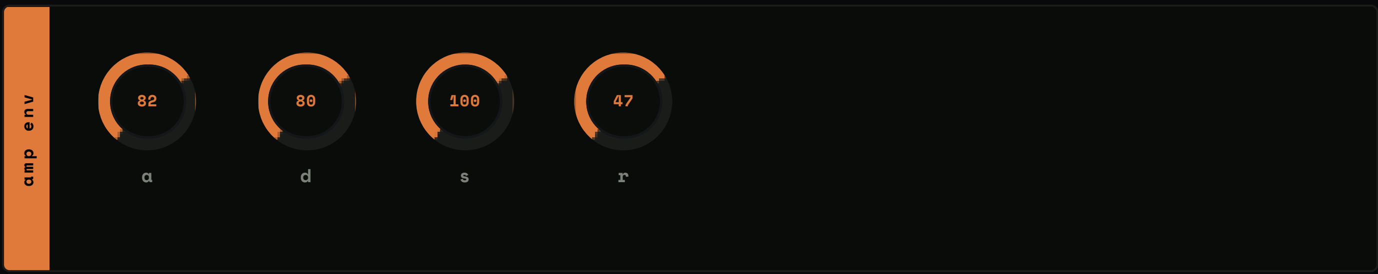
TWO-KNOB ENSEMBLE

DET around 0.15, SPRD at 0.7, on any of the saw-based engines (**8 DualDet**, **11 SawCloud**, **27 SuprSaw**) and the chord opens up like a window. Two knobs, done.

// 08 — THE AMP ENVELOPE

A, D, S, R.
Per voice. Independently.

Row 2 of the right block. Standard ADSR, one per voice, exponential curves throughout. The amp envelope drives each engine's level input — voices freed after the release tail fully decays. They don't sit around occupying slots.



KNOB	FUNCTION
A	Attack time, 1 ms → 4 s.
D	Decay time, 1 ms → 4 s.
S	Sustain level, 0–100 %.
R	Release time, 1 ms → 4 s.

REMEMBER

Short R on percussion engines (**21 BassDrum**, **22 Snare**, **23 HiHat**, **31 NsDrums**). Long R on the resonators (**32 Modal**, **33 SymStr**, **34 CombRes**) — they want to ring out into the next note.

THE SHAPE

Fig. 04 — Exponential curves throughout. One envelope per voice.

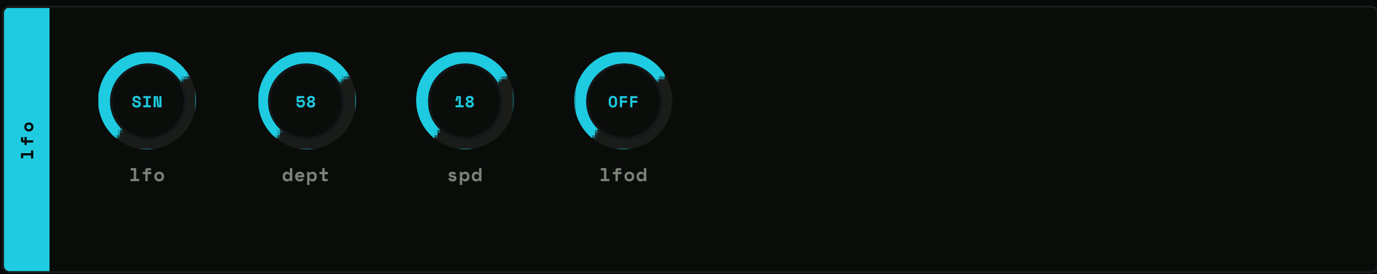
THE VOICE LOOP

Voices that have finished releasing skip rendering entirely. CPU stays where it needs to be. The allocator reuses freed voices before stealing — see the next page for the order.

// 09 — THE LFO

One LFO.
Six shapes. Seven destinations.

Row 3 of the right block. Single global LFO, but **each voice samples it at its own phase** — so a slow LFO routed to pitch produces gentle ensemble drift rather than a phase-locked wobble.



KNOB	FUNCTION
LFO	Waveform — Sin · Tri · Saw · Sqr · S&H · Rnd .
DEPT	Modulation depth, 0–100 %.
SPD	Rate, ~0.1 Hz → ~50 Hz, exponential.
LFOD	Destination — Off · Pitch · Timbr · Morph · Harm · Filt · Amp .

WHERE IT GOES

- **Pitch** → vibrato.
- **Timbr / Morph / Harm** → animated timbre — modulates the corresponding TIM / MOR / HAR input.
- **Filt** → cutoff wobble — sums into the filter envelope's modulation, doesn't replace it.
- **Amp** → tremolo.

SIX SHAPES

Fig. 05 — Five of the six shapes; **Rnd** is a smoothed sample-and-hold.

TWO QUICK DESTINATIONS

Sin / depth 8 % / speed 5 Hz / dest **Pitch** is classic vibrato. Sin / depth 30 % / speed 0.3 Hz / dest **Filt** is a slow filter sweep that breathes under a held pad.

More LFO destinations to come!

// 10 — VOICES, ALLOCATION, SIGNAL FLOW

Eight voices.
One signal path.

Under the hood, TRESSE is eight independent voices feeding a single stereo path. Knowing the order saves time when you can't figure out why a wavefold sounds too clean, or why a long filter release sits behind a short amp tail.

THE PATH, TOP TO BOTTOM

```
MIDI Note In
  │
  ├── Portamento  (per-voice glide from previous note)
  ├── Detune  (per-voice pitch spread)
  ▼
Engine render  (per voice, 8 voices)
  ▼
Stereo Mixer  (Spread panning) → Per-engine gain normalisation
  │
  ├── SEM Filter  (LP→BP→NCH→HP morph, Filter ADSR + LFO mod)
  ├── Wavefold  (post-filter)
  ▼
Soft-Clip Ceiling  (-0.3 dBFS, asymptotic)
  ▼
Stereo Out
```

VOICE ALLOCATION

When a note arrives, TRESSE picks a voice in this order:

01 Reuse the same MIDI note if it's already playing.

02 Find an inactive slot.

03 Grab a voice that's already in release.

04 Steal the oldest active voice.

Idle voices (no gate, envelope at zero) skip rendering entirely. CPU stays where it needs to be.

SAMPLE-RATE INDEPENDENCE

TRESSE renders internally at **48 kHz** and resamples to the host sample rate with a Lagrange interpolator. Pitch is identical at 44.1, 48, 88.2, and 96 kHz. Tune your synth at home, run a session at 96 — same patch, same sound.

HEADS UP

8 voices is the hard ceiling. The plugin steals oldest beyond that. If you're hearing notes drop out under a busy chord, you're stealing — back off the playing, or use sustain pedal sparingly.

// 11 — ENGINE REFERENCE

Thirty-nine ways to make a tone.

Every engine maps the four primary knobs differently. The notes below tell you what TIM / MOR / HAR / FM each one owns. Switching engines doesn't reroll the knobs — so you can A/B engines under the same parameter values and hear what each character brings.

01 CORE SOUND ENGINES

PART 1 · 13 OF 24

00 EastCoast
SUBTRACTIVE

Classic VCO → resonant filter. **TIM** = cutoff, **MOR** = resonance, **HAR** = waveform. The bread-and-butter analog voice — your default starting point.

01 PhaseDist
PHASE DISTORTION

Casio-era PD with two-stage shaping. **TIM** and **MOR** sculpt the distortion curve; **HAR** sets a second waveshape. Digital edge, clean fundamentals.

02 6-Op FM I
FM SYNTHESIS

Six-operator FM, patch bank 1. Bell, electric piano, brass territory. **TIM** = modulation index. **HAR** = patch select within the bank.

03 6-Op FM II
FM SYNTHESIS

Same engine, patch bank 2. Different algorithms, different families — wider envelopes, more bells.

04 6-Op FM III
FM SYNTHESIS

Bank 3. The harder, more digital end of the FM world — gritty leads, metallic textures.

05 Terrain
WAVE-TERRAIN

A pointer wanders over a 2D terrain map; eight terrains. **TIM** = X, **MOR** = Y, **HAR** = terrain select. Wide, evolving, hard to pin down.

06 StrChor
STRING MACHINE

Polyphonic string-machine emulation with built-in chorus. **TIM** = brightness, **MOR** = chorus depth. Solina territory.

07 Chiptune
8-BIT

NES-style square / triangle / noise voice with internal arpeggiator. **HAR** drives the arp; **FM** sets vibrato. Hold a chord, the arp does its work.

08 DualDet
DUAL VCO

Two detuned oscillators (saw / square / triangle). **TIM** = detune, **MOR** = waveform crossfade. Fat. Pairs with the DET knob for a triple-stack feel.

09 WavShape
WAVESHAPING

Sine through a polynomial shaper into a folder. **TIM** drives shape; **HAR** drives fold. Cuts through any mix without needing a filter to help.

10 PhaseMod
CLASSIC FM

Two operators with feedback, the canonical PM topology. **TIM** = ratio, **MOR** = feedback, **HAR** = depth. Smaller surface than 6-Op, faster to dial.

11 SawCloud
SAW STACK

Eight detuned saws with granular control over the stack. **TIM** = density, **MOR** = detune width. Thicker than a supersaw, looser at the edges.

12 Additive
ADDITIVE

Harmonic spectrum sculpted by movable "bumps". **TIM** = bump position, **MOR** = width, **HAR** = number of bumps. Vowel-like, organ-like, anything in between.

01 CORE SOUND ENGINES · CONTINUED

PART 2 · 11 OF 24

13 Wavetabl
WAVETABLE

Single-table scanning with optional lo-fi mode. **TIM** = position, **MOR** = lo-fi amount. Subtle at the low end, gnarly at the top.

14 Chords
CHORD GENERATOR

Plays a full chord from one MIDI note. **HAR** = chord type (major, minor, 7th, sus, dim, cluster…). **TIM** = inversion. Don't play your own chords on top — you'll voice-steal yourself.

15 Speech
FORMANT

Vowel and word synthesis. **TIM** = phoneme, **MOR** = speed, **HAR** = formant shift, **FM** = intonation. Sweep MOR slowly for the eerie effect.

16 Swarm
BUZZ CLUSTER

A swarm of buzzing oscillators, all slightly off. Wasps, hornets, broken radios. Pairs well with a wide filter sweep.

17 FiltNois
FILTERED NOISE

Noise through a resonant filter with an internal clock. **TIM** = cutoff, **MOR** = resonance, **HAR** = clock rate. Rhythmic textures without a sequencer.

18 Particle
DUST NOISE

Sparse impulses scattered through a small reverb. **TIM** = density, **MOR** = decay. Texture, not pitch — sits behind everything.

19 ResInh
INHARMONIC RESONATOR

A bank of inharmonic modes excited by the input. Glass, bowls, struck metal. Long releases reveal the partials.

20 ResMod
MODAL RESONATOR

Plate / bar / string / membrane modal model. **HAR** picks the body, **TIM** picks the brightness, **MOR** sets damping. Acoustic feel, no samples.

21 BassDrum
DRUM SYNTH

Analog-modeled kick. Short envelope, pitched sweep. Use a short D . **TIM** shapes the click; **HAR** tunes the body.

22 Snare
DRUM SYNTH

Analog-modeled snare. Noise plus a tuned shell. **MOR** sets the noise/tone balance; **HAR** tunes the shell.

23 HiHat
DRUM SYNTH

Analog-modeled metal hat. Short D , fast A . **TIM** sets brightness; **HAR** sets the metallic spread.

ENGINE NOTES THAT SAVE TIME

FM banks (2-4) load fixed patch data — some sound quiet by default; that's the source material, not a bug. Push FM for more harmonic content.
Chords (14) plays a full chord from one note — single key in, full voicing out.
Drums (21-23) want short D . Long sustain on percussion is rarely what you want.

02 TRESSE CUSTOM ENGINES

15 ENGINES · 24–38

Ported up from the Tresse v44 hardware firmware. These give the plugin its house character — most don't sound like anything you've heard from a Plaits-style box.

24 Karplus
PLUCKED STRING

Karplus-Strong with DC-blocked feedback. **TIM** = damping (bright when high), **HAR** = feedback (long ring near 1.0). Sweet spot lives between.

25 ByteBeat
ALGORITHMIC

Eight bytebeat formulas — the cult corner of synthesis. **HAR** picks the formula; **TIM** and **MOR** sculpt it. Digital, weird, alive.

26 CZ-PD
PHASE DISTORTION

Casio CZ-style PD oscillator. **TIM** drives the distortion waveform, **MOR** picks the second stage. Glassy attacks, hollow sustains.

27 SuprSaw
SUPERSAW

Seven detuned saws, PolyBLEP band-limited so high notes don't alias. **TIM** = detune, **MOR** = mix balance. Trance lead at half-mast, pad at full.

28 Formant
VOWEL SYNTHESIS

Pure vowel formants without the speech-engine overhead. **TIM** picks the vowel, **MOR** the formant width. Choral, throaty, surprisingly musical.

29 2-OpFM
FM

Stripped-down two-operator FM with feedback. The simplest possible FM voice, and probably the easiest to dial in. Try as a bell with long D .

30 WavFold
WAVEFOLDER

Multi-waveform input through a folder, PolyBLEP saw/square. **TIM** = fold amount, **MOR** = symmetry. Buchla west-coast tone.

31 NsDrums
NOISE DRUMS

Pitched noise drum with envelope. Build any kick, snare, or tom from one source. **HAR** sweeps the pitch envelope speed.

32 Modal
MODAL

8-partial modal resonator with struck and bowed excitation. Bell, bar, bowl, glass — ring until you tell it to stop. Use long R .

33 SymStr
SYMPATHETIC STRINGS

Three coupled strings, DC-blocked. They resonate against each other — play one note and the others ring along. Sustain-pedal piano in a box.

34 CombRes
COMB RESONATOR

Four-tap comb with struck and bowed modes. Pitched, percussive, woody. Marimba-adjacent at short decays; flute-adjacent at long.

35 Drawbars
ADDITIVE ORGAN

Hammond-style drawbars with tilt EQ and drive. **TIM** and **MOR** set the drawbar mix; **HAR** drives tilt. Add the LFO on amp for proper Leslie.

36 Complex
WEST COAST

Complex oscillator: FM into a phase-domain wavefolder. The Buchla side of the room. Inharmonic by default, lush at the edges.

37 Grains
GRANULAR

Pitched grain noise with smoothing. Sits between noise and tone. Useful for textures that need pitch but not melody.

38 FreqFM
MULTI-WAVE FM

Continuous FM ratio (0.5–8.0, allows inharmonic/metallic), morphing waveforms on carrier and modulator, mod-depth envelope. The deepest single engine in the box. Whole numbers = clean; irrational (1.41, 2.71) = metallic.

// 12 — PRESETS & MIDI

.tresse files. Standard MIDI.

The footer carries the preset name and the two buttons that read and write it. The DAW also persists the plug-in's state with the project — your patch comes back with the session whether you saved it or not.

PRESETS

Click `SAVE` to write every knob value plus the selected engine and name to a `.tresse` file. Click `LOAD` to read one. The DAW also persists the plug-in's state with the project — your patch comes back with the session whether you saved it or not.

Default preset name is `INIT`. Loaded preset names are uppercased and shown in the footer.

> **HEADS UP · THE BROWSER IS SETTLING**
>
> A proper in-window preset browser is coming in the next build. For now, use Finder + `LOAD`.

MIDI

- **Note On / Off** — Velocity drives the engine `level` input.
- **All Notes Off / All Sound Off** — Recognized; silences every voice immediately.
- **Pitch Bend / CC** — Not currently routed. Planned for the next build.

> **WHY MIDI IS MINIMAL (FOR NOW)**
>
> The plugin's surface is small enough that DAW host automation already covers most of what you'd map. MIDI Learn and Bend land in v1.1.

// 13 — THE OUTPUT STAGE

Minimal. Transparent.

There's no compressor. No makeup gain. No multiband anything. The output stage is three simple steps that exist only to keep you from clipping accidentally.

01 **Per-engine gain normalization.** Quiet engines run at 1.0×. Hotter customs (**ByteBeat 0.7×**, **SuprSaw 0.8×**, **Drawbars 0.7×**) get scaled down so engine swaps don't blow your ears off.

02 **The filter self-saturates.** No separate clamp needed. The TPT loop handles its own headroom.

03 **Soft-clip ceiling at -0.3 dBFS.** Asymptotic, branch-free. Transparent below threshold; smoothly rounds peaks above. Engages only on rare transient overshoots.

No "compressed" character. The filter and per-engine gain handle level discipline. The soft-clip is purely a safety net.

> **WHY THIS MATTERS**
>
> TRESSE is meant to play **cleanly** into your chain. Run a saturator after it if you want grit; run a bus compressor after that if you want glue. The plugin doesn't try to mix itself.

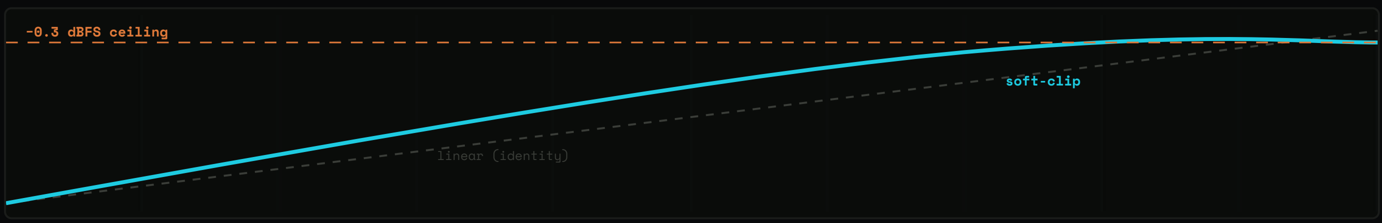


Fig. 06 — Linear (dashed) vs. the asymptotic ceiling. Below threshold the two are indistinguishable.

// 14 — HARDWARE → PLUGIN: WHAT CHANGED

What changed from the box.

For anyone arriving from the hardware Tresse — here's what's the same and what isn't. Most of the synthesis character carries over unchanged; the envelope and voice counts are where it gets generous.

FEATURE	HARDWARE (V44)	VST3
POLYPHONY	3 voices	**8 voices**
SAMPLE RATE	32 kHz	**48 kHz internal**, resampled to host
ENGINES	39	**39** · identical bank
FILTER	SEM-style continuous morph	TPT SEM-style — **cleaner, self-stable**
FILTER ADSR	A + D only	Full **A · D · S · R**
AMP ADSR	A + D only	Full **A · D · S · R**
PRESET SLOTS	128 + 512 + 512 (LittleFS)	`.tresse` files + DAW state
BLE MIDI / OLED / ENCODERS	Yes	N/A — your DAW owns the I/O

SAME ENGINES. SAME CHARACTER.

The synthesis code is a direct port. A patch that worked on the box works here — you can read a hardware patch sheet, dial in the same knobs, and land in the same place. The new headroom (eight voices, full ADSRs, higher sample rate) is on top of that, not in place of it.

WHAT TO EXPECT TO SOUND DIFFERENT

The **filter** is cleaner — the v44 op-amp warmth isn't there. Add a saturator after the plugin if you miss it.
The **envelopes** have an S and R that the hardware didn't — most factory patches use them.
The **voice spread** is wider because you've got more voices to spread.

// 15 — FIVE WAYS TO START A PATCH

Patches you can dial in in thirty seconds.

Not presets — recipes. Specific knob settings that put you somewhere musical fast, with notes on what to break first.

// 01 — CLASSIC ANALOG LEAD

- Engine: **EastCoast** (0).
- TIM 0.7 · MOR 0.4 · HAR 0.3 · FM 0.
- TYPE 0 (LP) · CUT 0.6 · RES 0.6 · ENV 0.5.
- fA 0 · fD 0.3 · fS 0.2 · fR 0.4.
- A 0 · D 0.4 · S 0.8 · R 0.3.
- DET 0.1 · SPRD 0.5 · PORT 0.15 · WAVF 0.

Plucky filter open on attack, sustains, glides slightly between notes. Push RES up for more bite.

// 02 — HELD ENSEMBLE PAD

- Engine: **SuprSaw** (27).
- TIM 0.6 · MOR 0.5 · HAR 0.3 · FM 0.
- TYPE 0.1 (LP-ish) · CUT 0.5 · RES 0.3 · ENV 0.4.
- fA 0.6 · fD 0.5 · fS 0.7 · fR 0.7.
- A 0.5 · D 0.3 · S 0.9 · R 0.7.
- DET 0.18 · SPRD 0.85 · PORT 0 · WAVF 0.
- LFO Sin · DEPT 0.25 · SPD 0.15 · LFOD **Filt**.

Slow filter sweep under a wide, detuned saw stack. Hold a chord and let it breathe.

// 03 — FM BELL

- Engine: **2-OpFM** (29) or **FreaqFM** (38).
- TIM 0.55 · MOR 0.3 · HAR 0.4 · FM 0.7.
- TYPE 0 (LP) · CUT 0.85 · RES 0.1 · ENV 0.
- fA 0 · fD 0 · fS 1.0 · fR 0 (filter env off).
- A 0 · D 0.7 · S 0 · R 0.6.
- DET 0 · SPRD 0.3 · PORT 0 · WAVF 0.

Percussive attack, long decay. Open the filter all the way — the bell shape lives in the engine. Sweep HAR to retune the partials.

// 04 — KARPLUS PLUCK INTO RESONANT FILTER

- Engine: **Karplus** (24).
- TIM 0.4 · MOR 0.5 · HAR 0.85 · FM 0.
- TYPE 0.1 · CUT 0.6 · RES 0.65 · ENV 0.6.
- fA 0 · fD 0.5 · fS 0.1 · fR 0.4.
- A 0 · D 0.05 · S 1.0 · R 0.3.
- DET 0.05 · SPRD 0.4 · PORT 0 · WAVF 0.
- LFO Tri · DEPT 0.15 · SPD 0.4 · LFOD **Filt**.

A picked string with a wet, resonant filter shimmer on each note. Plays well with a delay after the plugin.

// 15 — CONTINUED

One more recipe. Then the loop.

Use these as starting points — never as destinations. Once any of them lands somewhere good, the next move is RANDOM on whatever isn't carrying the patch.

// 05 — DRONE / TEXTURE

- Engine: **Modal** (32) or **SymStr** (33).
- TIM 0.5 · MOR 0.7 · HAR 0.6 · FM 0.
- TYPE 0.3 (BP) · CUT 0.5 · RES 0.4 · ENV 0.
- fA 0.4 · fD 0.5 · fS 0.5 · fR 0.6.
- A 0.3 · D 0.5 · S 0.8 · R 1.5.
- DET 0.2 · SPRD 0.9 · PORT 0.3 · WAVF 0.2.
- LFO Sin · DEPT 0.3 · SPD 0.05 · LFOD **Morph**.

Resonators ringing into each other under a slow morph modulation. Play three notes, hold them, walk away.

WHAT "BREAK FIRST" MEANS

Each recipe has one knob whose value carries the character — RES on the lead, SPRD on the pad, HAR on the bell. That's the knob to leave alone. Reroll the rest.

THE RANDOM LOOP

Once any recipe lands somewhere good, the next move is RANDOM until it falls apart, then back the knobs out until it returns. That's how you learn the engine. The plugin is built around this rhythm.

A PRACTICAL SESSION

Hold a chord. Load recipe **02**. Tap RANDOM five times — listen to each. Lock the one you like by tapping SAVE . Now load recipe **04** and repeat. You'll have ten patches in ten minutes that all sound like **you**.

ONE THING TO REMEMBER

The engine name is the only thing RANDOM never touches. If the patch isn't going where you want, swap the engine first — then reroll.

// FIN

Tweak an engine or a random patch. Until it's good.

That's most of the workflow. The plugin gets out of the way and lets the engines speak. If something here is unclear — or if you'd like a new engine in a future build — write us. We read everything.

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